21ASC305T
ROCKET PROPULSION

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- ***Unit-1 - Basic Concepts 9 Hour***
- *Rocket propulsion systems -working principle, Classification of rockets, Rocket equation, Rocket Nozzles and its classifications -Nozzle performance- equilibrium and frozen flow, Characteristic velocity and Thrust coefficient, Performance parameters and efficiencies of rocket, Staging and clustering of rocket*
- ***Unit-2 - Chemical Propellants 9 Hour***
- *Chemical propellants -Molecular mass, Specific heat capacities, Specific heat ratio, mixture ratio, Stoichiometric ratio and equivalence ratio, Energy release during combustion-Heat of formation and Heat of combustion-Combustion Instabilities-Criterion for choice of propellant, Solid Propellants-Composition and processing, Liquid Propellants-Classifications of liquid propellant, Storability of liquid propellant, Chemical Equilibrium*
- ***Unit-3 - Solid Rocket Engine 9 Hour***
- *Propulsion Elements for Solid Rocket Motors, Solid propellant combustion and Internal Ballistics of Motors -Mechanism of burning, Ignition System-Igniter types - Solid propellant grain design - Burn Rate-Factors influencing burn rates, burn rate index for stable operation, Action time and burn time, Design of Solid Propellant rocket, Simulation of solid rocket motor performance.*
- ***Unit-4 - Liquid Rocket Engine 9 Hour***
- *Liquid Propellant Rocket-Hardware components and its functions, Propellant feed systems –Gas pressure feed system, Turbo pump feed systems and engine cycles, Tank Pressurization - Types of injectors – Thrust chamber, Cooling of Thrust chamber - Cryogenic propulsion system and its special features, Simulation of Liquid rocket performance, Hybrid rockets- Review of solid-fuel regression rate behavior in classical and non-classical Hybrid Rocket Motors - Rocket engine design using software.*
- ***Unit-5 - Non-conventional Propulsion Techniques 9 Hour***
- *Non - conventional propulsion techniques -Electrical Rockets-Electro-thermal propulsion system-Arc-jet thruster, Resistojet Thruster-Electro-static thruster-Ion thruster, Bombardment Ionization, Hall Effect Thruster -Electro-magnetic propulsion system-Magneto plasma dynamic accelerator, Pulsed Plasma accelerator- Laser Propulsion -Nuclear powered rockets- Solar Propulsion system*

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- *1. George P. Sutton, Oscar Biblarz, "Rocket propulsion elements", Wiley India Pvt Ltd. Ninth Edition, 2017*
- *2. Ramamurthi.K, "Rocket propulsion", Laxmi Publications, India, Second edition, 2019.*
- *3. J D. Mattingly and K M. Boyer,"Elements of Propulsion: Gas Turbines and Rockets", Second Edition,2016*
- *4. Philip Hill and Carl Peterson, "Mechanics and thermodynamics of propulsion", Pearson India, second edition, 2010.*
- *5. Stephen R. Turns, "An Introduction to Combustion: Concepts", McGraw-Hill Education; 4th edition, 2020*
- *6. Yahya S M, "Gas Tables",8th Edition, 2018*

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Unit-3 - Solid Rocket Engine

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Burning Rate Relation with Pressure

- With many propellants it is possible to approximate the *burning rate* as a function of *chamber pressure*, at least over a limited range of chamber pressures.

- For a majority of production-type propellants the most commonly used empirical equation is
  - $r = aP_1^n$

- when $n > 1.0$, any pressure disturbances present will be amplified in the chamber.

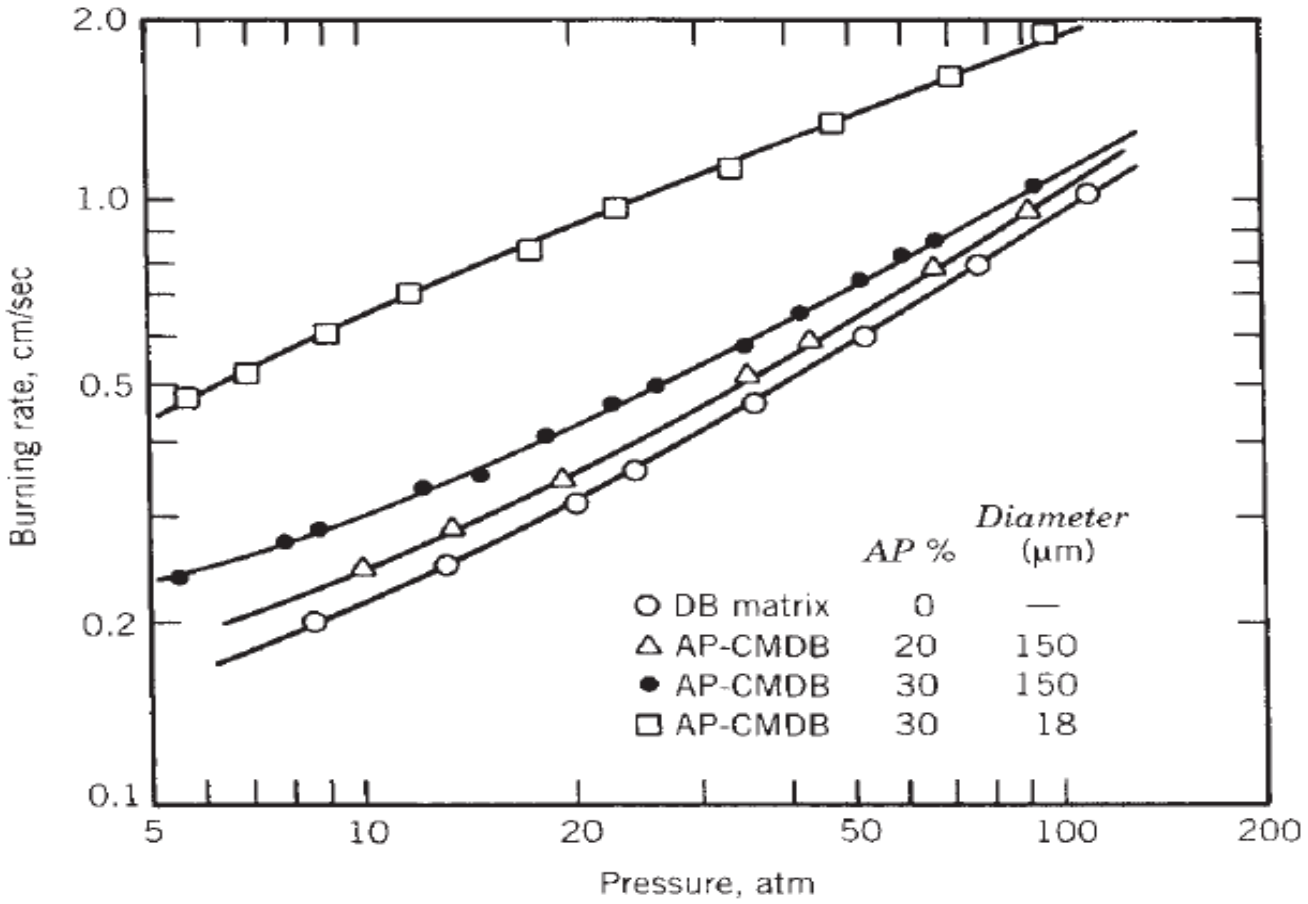
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Burning Rate Relation with Pressure

- For any particular propellant and for wide temperature and pressure limits, the burning rate can vary by factors of 3 or 4.
- For all propellants this translates to a range from about 0.05 to 75 mm/sec or 0.02 to 3 in./sec.
- To achieve such rates, combinations of very small sized ammonium perchlorate (AP), burning rate catalysts, additives, or embedded metal wire are needed.

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When the size of the AP particles is reduced or the percentage of AP is increased, an increase in burning rate is observed



Burning Rate Relation with Pressure

- High values on *n* give rapid changes of burning rate with pressure.
- Even an ordinarily small variation in chamber pressure can induce substantial changes in the amount of hot gas produced.
- Most commercially available propellants have a pressure exponent *n* ranging between 0.2 and 0.6.

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Burning Rate Relation with Pressure

- Common practice in developing and testing larger rocket motors is to "condition" the motor for many hours at a particular temperature before ground-test firing it to ensure that the propellant grain is uniformly at the desired temperature since rocket motor performance characteristics must stay within specified acceptable limits.

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Burning Rate Relation with Pressure

- As ***n*** **approaches 1.0**, the burning rate and chamber pressure become very sensitive to one another and a disastrous rise in chamber pressure may occur in only a few milliseconds.

- ***Plateau propellants*** exhibit nearly constant burning rate over a limited pressure range, and they are desirable for minimizing effects of changes in initial temperature on motor operation.

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Burning Rate Relation with Ambient Temperature (T_b)

- Several double-based propellants and a few composite propellants are known have this desirable plateau characteristics.

- For air-launched missile motors, the extremes are usually 219 K (−65◦F) and 344 K (160◦F) and for silo or submarine launched motors 266 K (20◦F) to 300 K (80◦F).

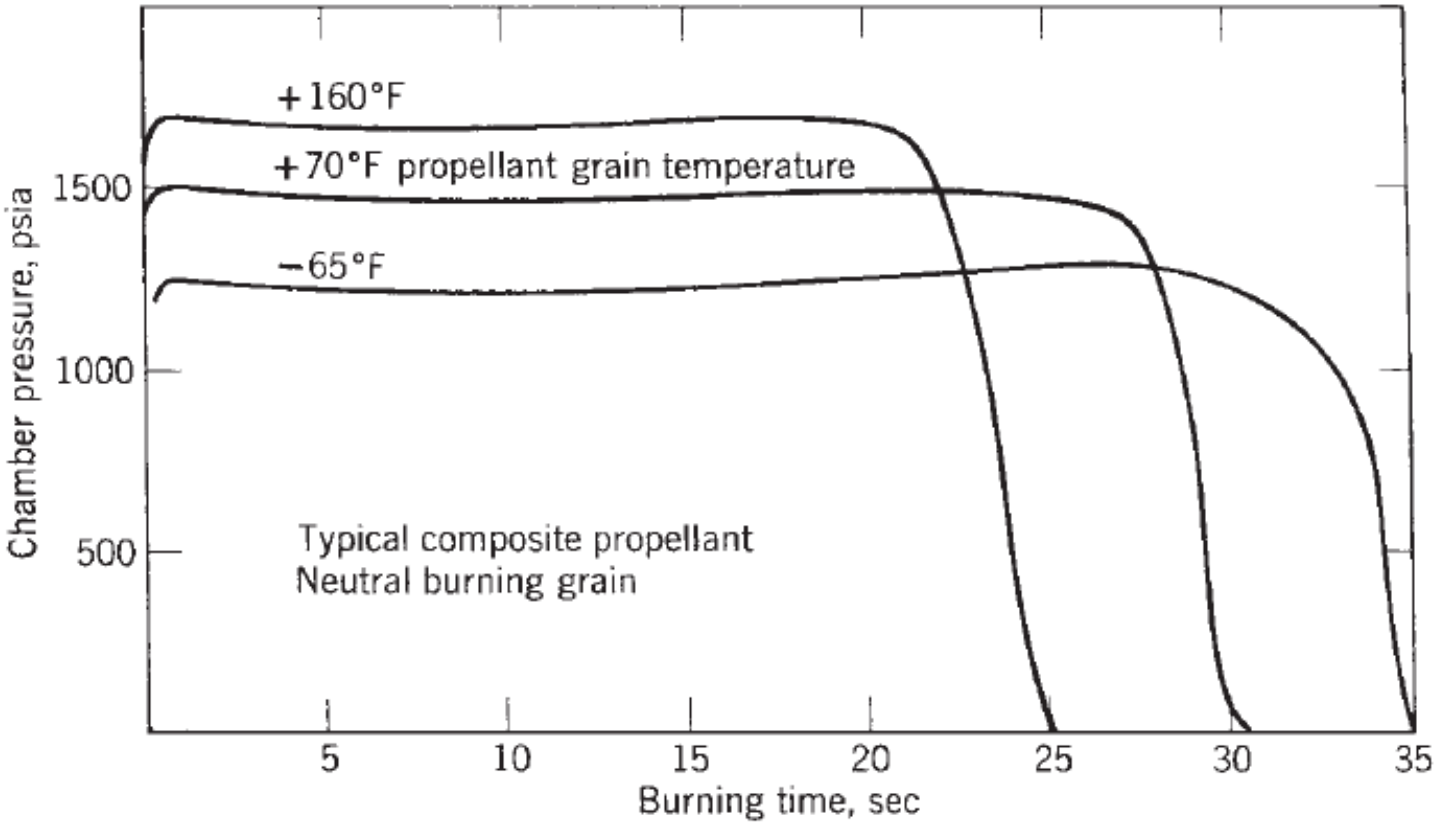
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Burning Rate Relation with Ambient Temperature (T_b)

- Motors using typical composite propellant mixtures may experience a 20 to 35% variation in chamber pressure and a 20 to 30% variation in operating time over such a range of propellant temperatures

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Effect of propellant initial temperature on burning time and chamber pressure



Burning Rate Relation with Ambient Temperature (T_b)

- The sensitivity of the burning rate to propellant temperature can be expressed in the form of temperature coefficients, the two most common being

$$K \equiv A_b/A_t$$

$$\sigma_p = \left(\frac{\partial \ln r}{\partial T_b}\right)_{p_1} = \frac{1}{r}\left(\frac{\partial r}{\partial T_b}\right)_{p_1}$$

$$\pi_K = \left(\frac{\partial \ln p_1}{\partial T_b}\right)_K = \frac{1}{p_1}\left(\frac{\partial p_1}{\partial T_b}\right)_K$$

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Burning Rate Relation with Ambient Temperature (T_b)

- with σ_p known as the *temperature sensitivity of burning rate*, expressed as the change of burning rate per degree change in propellant ambient temperature at a fixed value of chamber pressure, and

- π_K known as the *temperature sensitivity of pressure* expressed as the change of chamber pressure per degree change of propellant ambient temperature at a particular value of *K*.

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Variable Burning Rate Exponent n

- Burning rate data indicates that n may not be really constant but a function of p_1 (which indirectly makes it also a function of T_b).

- It may no longer be possible to accurately predict the pressure excursions with ambient temperature changes; that in the constant-n case π_K depends only on a material/ propellant property (σ_p) and on n.

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Burning Enhancement by Erosion

- ***Erosive burning*** refers to any increase in the propellant burning rate caused by the high-velocity flow of combustion gases across the burning propellant surface.

- It can seriously affect the performance of solid propellant rocket motors.

- It occurs primarily in port passages or in grain perforations as the combustion gases flow toward the nozzle;

- it is more likely to occur when the port passage cross-sectional area A is not large relative to the throat area A_t, with a port-to-throat area ratio of 4 or less.

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